

Hybrid Learning against Label Noise: Combining Contrastive Pretraining and Structural Label Regularization.

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ABSTRACT Cross-entropy deep networks can generalize successfully in clean supervision, although they overfit and memorize corrupted labels in the event that the targets are noisy. This is more so when it concerns the labels assigned by human beings where disagreement is normally as a result of ambiguity rather than random change in the labels. As presented in this paper, we take practical robustness methods on CIFAR-10 (clean labels) and CIFAR-10N (human labels) and evaluate them using a fixed ResNet-18 training pipeline (SGD, 100 epochs, identical optimization and augmentations). We compare four CIFAR-10 versions (30 percent subset, CLEAN), (i) Baseline cross-entropy classifier, (ii) Structural method that incorporates neighbor-consistency regularization, (iii) Hybrid method that incorporates neighbor-consistency regularization and agreement-based sample reweighting, (iv) Contrastive-init variant, which uses contrastive-style initialization and contrastive-style pretraining. We report Baseline, Structural, and Hybrid results on CIFAR-10N (30 percent subset, AGGRE) (Contrastive-init has not been tested and evaluated on CIFAR-10N in the current experimental set and is considered a clean-label diagnosis). Structural achieves the best peak validation accuracy (0.7873 at epoch 54) compared to Hybrid (0.7627), Baseline (0.7573), and Contrastive-init (0.7340). Structural has the highest peak on CIFAR-10N (30 percent AGGRE) (0.8753 at epoch 76) whereas Hybrid is stable at late training (0.8664 at epoch 100). Besides aggregate accuracy, we analyze learning curves and confusion matrix and embedding visualizations (UMAP and t-SNE) to learn how classes act and representation geometry. Overall, the results can be summed up as the conclusion that structurally consistent local prediction improves training recipe resilience to human label noise, which is also in line with the past results that suggested that structure can mitigate overfitting to corrupted supervision [1], [3].

INDEX TERMS Label noise, CIFAR-10N, robustness, neighbor consistency, resampling weights, contrastive initialization, ResNet-18, UMAP, t-SNE.

I. INTRODUCTION

Label noise is a realistic and challenging supervised learning problem. Image benchmarks that are commonly used contain ambiguous instances and the occurrence of annotation errors occasionally due to occlusion, low-resolution, category confusions or genuine uncertainty. It is among the failure modes that standard cross-entropy training has it will quickly learn simple patterns but will subsequently overfit harder or mislabeled ones, then subsequently memorize noise and harm generalization.

Of special interest is human label noise which need not necessarily be synthetic and need not necessarily be uniformly random. The disagreement among annotators is often an indication of the ambiguity in the image content and not

the ludicrousness of labels changing. CIFAR-10N provides one with a real-world setting: human labeling of CIFAR-10 images using labels in different variants of noise [6]. In this paper we focus on the AGGRE label set that is a combination of human judgment and a challenging but realistic supervision signal.

As part of the knowledge gap, the robustness techniques can be exercised in different architectures, schedules and training requirements, thus it is difficult to extrapolate the performance variability on the robustness mechanism itself. Our goal is to reduce these confounds by training across different mechanisms with the help of a single training pipeline.

The research questions studied are as follows:

RQ1: Does neighbor-consistency regularization perform

better under presence of human label noise (CIFAR-10N AGGRE) than a baseline based on cross-entropy?

RQ2: Does agreement-based reweighting (Hybrid) with neighbor-consistency provide superior peak accuracy and/or late-epoch stability to human label noise?

RQ3: What are the relationships between neighborhood constraints and classwise confusions and geometry of learned representations in UMAP/t-SNE?

II. RELATED WORK

Learning with label noise techniques are resilient learning to label correction, sample selection, co-teaching, semi-supervised learning and consistency-based regularization. The other theme that continues to recur is that the dynamics of representation learning are also tightly connected to the notion of robustness. In a case where features remain constant, and semantically valid, the probability of models memorizing particular corrupted labels is low.

Contrastive learning has been researched as a representation learning algorithm that is not as contingent on possibly erroneous indicators. Ghosh et al. state that contrastive objectives are advantageous to increase robustness in the case that there is a label noise and the semantic similarity is emphasized, rather than on the noisiness of labels [1]. SimCLR is a paradigm model that promotes contrastive pretraining, followed by classification [2]. In contrast to contrastive-only methods, our paper is based on the contrastive-only approach, which is why this approach is an illustration of a clean-label diagnostic, and consistency to the neighborhood is the primary robustness condition under human noise.

The second option is to add more structure than one-hot targets. According to Kim et al., one can learn through structural labels and they indicate that more informative signals of supervision can boost learning with noisy labels [3]. We do not replicate their entire form with feature structures.

III. METHODS

A. PROBLEM SETUP

Classifier $f_\theta(x)$ that predicts the logits of $Z_\theta = z(x)$ in $C = 10$ classes in $\mathbb{R}^C$ is trained. The predicted probability of class c is:

$$p(y = c|x) = \frac{\exp(z_c)}{\sum_{j=1}^C \exp(z_j)} \quad (1)$$

When dealing with CIFAR-10 (CLEAN) we train on regular dataset labels. For CIFAR-10N (AGGRE), we are trained on human-provided AGGRE labels, and test on clean CIFAR-10 test labels, the CIFAR-N protocol [6]. We present validation cross-entropy and validation accuracy per epoch in our report, and give a summary of each method including its maximum validation accuracy (and the number of epochs) and its accuracy at its final epoch 100.

B. BASELINE: CROSS-ENTROPY

Cross-entropy loss of a sample, $(x_i, y_i) =: CEL = (1/N) \sum \mathcal{L}_{CE}(x_i, y_i) = -\log p(y_i|x_i)$. Under the label

noise category, one can use the observed labels as ground truth, which causes the model to fit incorrect targets late in training and memorizing more.

C. NEIGHBOR-CONSISTENCY STRUCTURAL

Structural method has a neighbor-consistency term that facilitates local smoothness in prediction space. Where h_i is the penultimate-layer feature of sample x_i . To each sample in the feature space we compute a set of K nearest neighbors, referred to as $\mathcal{N}_K(i)$. The neighbor-consistency loss is used to encourage the logits of a point to be close to the average logits of the neighbors:

$$\bar{z}_i = \frac{1}{K} \sum_{j \in \mathcal{N}_K(i)} z_j \quad (2)$$

$$\mathcal{L}_{NC} = \frac{1}{N} \sum_{i=1}^N \|z_i - \bar{z}_i\|_2^2 \quad (3)$$

The Structural objective is overall:

$$\mathcal{L}_{Structural} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{NC} \quad (4)$$

at the value of the $\lambda = 0.3$ in our experiments. Intuition. Isolated label errors often do not conform to the local semantic neighborhood of a sample, in general. Local consistency can be enforced and therefore it can be used to mitigate tendencies to fit idiosyncratic mislabeled samples.

D. HYBRID: NEIGHBOR CONSISTENCY + AGREEMENT-WEIGHTED CROSS-ENTROPY

Hybrid is a consensus of neighbor-consistency and an agreement-based reweighting rule. It is also a neighborhood agreement of our heuristic protecting suspicious samples and the weight of the sample w_i where w_i uses a weighting factor $\alpha = 0.5$. Formally, down weighting of a sample is achieved by having a sample whose prediction is inconsistent with local neighborhood predictions with a threshold of confidence of 0.6 (we take this to be 0.6).

The weighted cross-entropy is:

$$\mathcal{L}_{WCE} = \frac{1}{N} \sum w_i \mathcal{L}_{CE} \quad (5)$$

The Hybrid objective is:

$$\mathcal{L}_{Hybrid} = \mathcal{L}_{WCE} + \lambda \mathcal{L}_{NC} \quad (6)$$

E. CONTRASTIVE-INIT VARIANT: CONTRASTIVE INITIALIZATION + SUPERVISED TRAINING

The Contrastive-init variant is a representation learning objective that trains the encoder by contrastive style and finally is supervised by cross-entropy. Supervised training makes no use of any contrastive goal, and therefore any improvement is considered an effect of the initialization of supervised training.

As part of the evaluation scope, It is evaluated only on CIFAR-10 (clean) but not CIFAR-10N (AGGRE). We do not do this by deriving empirical robustness Contrastive-init claims in the situation of human label noise.

IV. EXPERIMENTAL SETUP

A. OPTIMIZATION, ARCHITECTURE AND TRAINING PROTOCOL

The experiments are each founded on ResNet-18 scaled to the CIFAR resolution. SGD is trained with 0.9 momentum, 5×10^{-4} weight decay, 0.1 learning rate, batch size 128 and 100 epochs. We are not prematurely halting to check on any potential late-epoch noise-degradation.

B. DATA AUGMENTATION AND DATA NORMALIZATION

Random crop is used on CIFAR-sized images, tensor conversion and normalization. Validation/test only involves conversion and normalization of tensors.

C. DATASETS, LABEL SOURCES, AND SPLIT CONSISTENCY

CIFAR-10 contains clean labels, as well as a withheld validation split. CIFAR-10N entails labels which are given by human beings; we are interested in AGGRE. Therefore, CIFAR-10N protocol is used to assess testing with Clean CIFAR-10 test labels [6]. We use the same protocol in every dataset and hence the difference is attributed to the mechanism of robustness and not the change in optimization or augmentation.

D. SUBSET MODE AND COMPUTE ENVIRONMENT

We use the fixed protocol which uses a training subset of 30 per cent when the subset mode is turned on to ensure feasibility under limited compute. The tests were conducted in the local area on an Apple Silicon laptop with a unified memory of 24 GB.

E. PLANNED NOISE-RATE SWEEP (EXTENSION)

CIFAR-10N has real human noise, though it does not disclose continuous control of noise-rate. One can build it by sweeping in controlled index alignment at proportion p with clean CIFAR-10 labels co-mixed with CIFAR10 N labels and degradation measured as a function of p [6].

V. RESULTS

A. PART A: SUMMARY OF PERFORMANCE

Since noisy-label training may improve and then deteriorate because of memorization, we indicate both peak and final (epoch-100) validation accuracy. Table 1 gives a summary of findings of CIFAR-10 (clean) versus CIFAR-10N (AGGRE).

TABLE 1. Comparative overall performance (peak validation accuracy, peak epoch, and final validation accuracy at epoch 100).

Section	Model	Peak acc	Epoch	Acc@100
CIFAR-10	Baseline	0.7573	50	0.6633
CIFAR-10	Structural	0.7873	54	0.7660
CIFAR-10	Hybrid	0.7627	99	0.7200
CIFAR-10N	Cont-init	0.7340	73	0.6393
CIFAR-10N	Baseline	0.8691	79	0.8452
CIFAR-10N	Structural	0.8753	76	0.8423
CIFAR-10N	Hybrid	0.8664	100	0.8664

Structural achieves the best peak performance (0.7660) and best peak (0.7873) on CIFAR-10 (clean subset) and it can be seen that the Structural neighbor-consistency regularization does not just improve it in the presence of noise, but can also stabilize learning with clean supervision. Baseline and Contrastive-init, on the other hand, exhibit greater decay between peak and epoch 100.

Structural has the highest peak (0.8753) on CIFAR-10N (AGGRE) a bit higher than Baseline (0.8691). Nevertheless, Baseline and Structural show a reduction at epoch 100, implying that they degrade at late epochs when trained on noisy human labels. Hybrid does not reach the highest peak, although it is stable, reaching its last value (peak) of 0.8664 at epoch 100.

B. PART B: LEARNING CURVES ON CIFAR-10N

Figures 1 and 2 indicate validation accuracy curves and cross-entropy of Baseline, Structural, and Hybrid on CIFAR-10N (AGGRE) respectively.

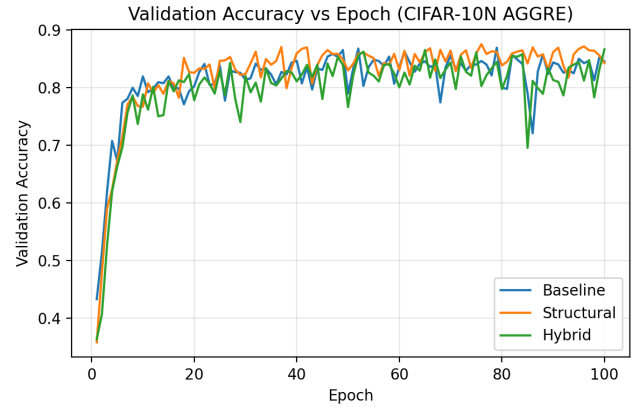


FIGURE 1. Validation accuracy vs. epoch on CIFAR-10N (AGGRE).

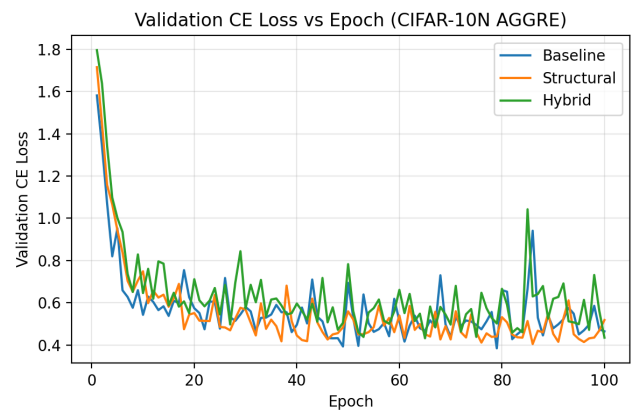


FIGURE 2. Validation cross-entropy vs. epoch on CIFAR-10N (AGGRE).

One of the patterns is the separation of peak and late epoch behavior. Baseline has a peak value of 0.8691 at epoch 79 and Structural has a peak value of 0.8753 at epoch 76 and drops

to 0.8423 at epoch 100. This peak-then-decline behavior is congruent with memorization dynamics.

Hybrid is not the same, therefore it has a higher peak accuracy at epoch 100 (0.8664) and does not degrade in late epochs in the reported case.

C. PART C: MATRIX OF CONFUSION AND CLASS-WISE BEHAVIOR

Figure 3 provides the confusion matrix of Structural model on CIFAR-10N (AGGRE). Table 2 provides class-based accuracies.

TABLE 2. Structural model Class-wise performance on CIFAR-10N (AGGRE) test set.

Class	Accuracy
0	0.8850
1	0.9480
2	0.8730
3	0.7930
4	0.9420
5	0.7330
6	0.9100
7	0.7960
8	0.9480
9	0.9250

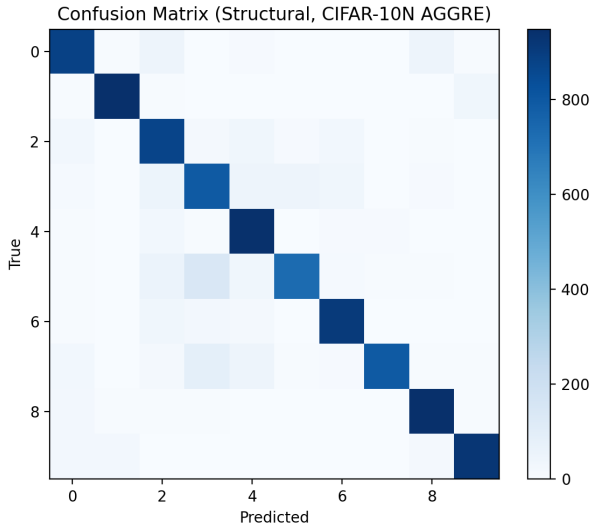


FIGURE 3. Confusion matrix Structural model on CIFAR-10N (AGGRE) test set.

The misclassifications are concentrated in visually adjacency pair animal classes (dog, cat, horse). The performance across the categories is not the same; Class 5 (0.733) has the lowest accuracy.

D. PART D: VISUALIZING (UMAP AND T-SNE)

The penultimate-layer representations of Structural and Hybrid are visualized in Figures 4-6.

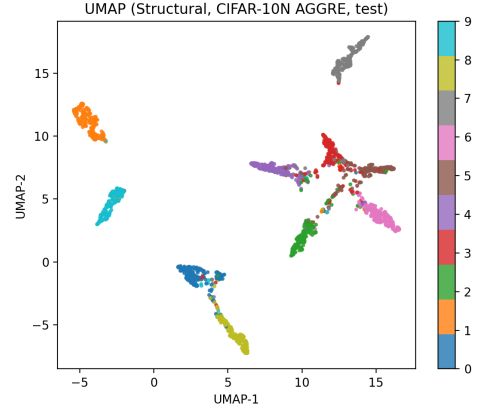


FIGURE 4. UMAP of penultimate-layer feature of Structural model on CIFAR-10N (AGGRE).

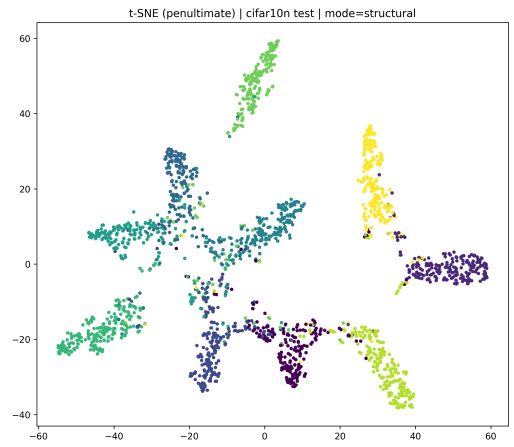


FIGURE 5. t-SNE of the penultimate-layer features of Structural model on CIFAR-10N (AGGRE).

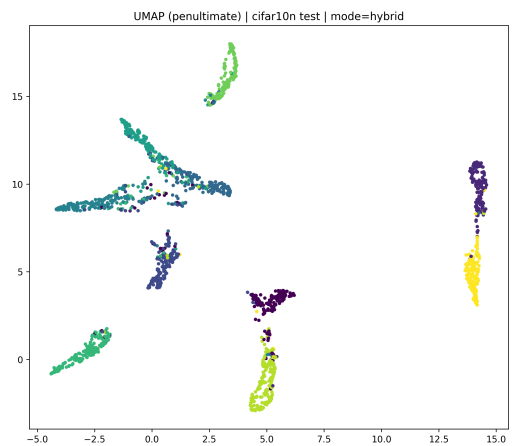


FIGURE 6. UMAP of penultimate-layer features of Hybrid model on CIFAR-10N (AGGRE).

Though both UMAP and t-SNE are qualitative tools, they assist in the interpretation of the effects of neighborhood constraints. This is visually represented by relatively stricter

within-class groupings and less fragmentation in a large proportion of the classes in Structural.

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VI. DISCUSSION

A. RESPONSES TO RESEARCH QUESTIONS

Human noise neighbor-consistency (RQ1). Structural neighbor-consistency achieves the highest validation accuracy (0.8753) which, however, is marginally better than the Baseline (0.8691), supporting the hypothesis that local consistency can be introduced to the training pipeline to reach greater resilience to human label noise.

RQ2 (Hybrid peak vs stability). Hybrid is not the Structural with best performance in AGGRE, however it is the one that is most stable in the late-epochs (0.8664 at epoch 100 with no decay). This indicates that even when the property of downweighting in memorizing late epochs is not in optimum form, agreement-based downweighting can be used to reduce the effect.

RQ3 (class behavior and geometry). Class-wise accuracy by class and confusion matrix indicate that the errors remaining are clustered between overlapping categories of animals and that is in line with the opinion that human errors are true ambiguity. The embedding visualizations provide additional qualitative data that regularization of neighborhoods encourages coherent local structure but the ambiguous classes are not fully mixed.

B. DYNAMICS OF MEMORIZATION INTERPRETATION

Information under label noise will be the difference between peak and epoch-100 performance. Both Baseline and Structural have a peak then declining dynamics on CIFAR-10N and this also fits in memorization dynamics: the longer the training process, the more it is likely to overfit to noisy labels, and the less likely to overfit to clean evaluation labels [6]. This drop is also minimized by hybrid, pointing to the idea that the downweighting of samples which are locally inconsistent can be an effective human noise stabilization heuristic.

C. CONTRASTIVE-INIT AS A CLEAN-LABEL DIAGNOSTIC

We add Contrastive-init as a clean-label diagnostic to study the hypothesis that contrastive-style initialization enhances downstream supervised training on the same schedule. In the existing experimental paradigm, Contrastive-init does not perform better than Structural on CIFAR-10, and is not

tested on CIFAR10 N. Hence, we do not take these results to demonstrate and/or refute contrastive robustness in the presence of human label noise. It is consistent in the literature that contrastive learning is resistant to label noise in certain regimes [1], but to confirm that statement in our pipeline, explicit noisy-label contrastive evaluation is necessary.

D. RELEVANCE TO STRUCTURAL SUPERVISION

We find that our result is consistent with the general principle of adding structure to enhance learning in the label noise condition. According to Kim et al., structural supervision can enhance learning in case unreliable labels are used [3]. Though we do not apply structural labels in practice, our neighbor-consistency regularization is a lightweight structural constraint which is used to stabilize training by using neighborhood relations in representation space.

VII. LIMITATIONS

First, single-seed results are mainly used; more powerful arguments would need so-called multiple-seed results with an average and standard deviation. Second, our paper lacks a sensitivity analysis of key parameter robustness of K , λ , and the level of agreement. Third, Contrastive-init is not tested on CIFAR-10N (AGGRE) in the existing experimental pool, and, as a result, we do not make empirical assertions regarding contrastive robustness to human label noise. Lastly, the suggested noise-rate sweep is provided as an extension but not actualized in the documents of results.

VIII. CONCLUSION

We conducted a controlled comparison of practical robustness strategies to learning with label noise on a constant ResNet-18 training pipeline on CIFAR-10 (clean) and CIFAR-10N (AGGRE human labels). Structural neighbor-consistency regularization had the highest peak accuracy with clean and human-noisy supervision, which is consistent with the hypothesis that local prediction smoothness can alleviate overfitting to corrupted supervision. Hybrid demonstrated the strongest late-epoch stability under AGGRE; hybrid reweighting was no longer tested as the optimal at its peak under AGGRE. Learning curve, confusion matrix and embedding visualization diagnostic analyses suggest that the remaining errors are concentrated in the naturally ambiguous classes and that neighborhood regularization fosters local coherent structure in the representation space. A natural next step is to test Contrastive-init on CIFAR-10N and to perform a noise-rate sweep by adding clean and human labels to each other in different ratios.

CONTRIBUTIONS

The project adds a confined assessment of lightweight label-noise robustness frameworks with one unchangeable instruction pipeline. We use and evaluate Baseline cross-entropy, Structural neighbor-consistency regularization, Hybrid neighbor-consistency and agreement-based reweighting, and a Contrastive-init clean-label diagnostic variant.

We report peak and epoch-100 performance to capture the dynamics of memorization in noisy supervision, and we conduct diagnostic analysis in terms of learning curves, confusion matrices inspection, accuracy reporting by classes, and representation visualizations (UMAP/t-SNE) to achieve interpretability and reproducibility.

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APPENDIX A INFORMATION ABOUT REPRODUCIBILITY

Reproducibility: The appendix replicates the information presented in the article.

Each experiment consists of a fixed ResNet-18 backbone and a unified training script with method flags of Baseline, Structural, Hybrid, and Contrastive-init. CIFAR-10N label sets are obtained from CIFAR-10N [6]; in particular, we use the AGGRE set of labels and load label files and mappings through the tools of the UCSC-REAL repository [7]. The CIFAR-10N protocol is evaluated on clean CIFAR-10 test labels [6]. Embedding visualizations are based on penultimate-layer features mapped to 2D with UMAP or t-SNE [4], [5].

APPENDIX B IMPORTANT LINKS

Project repository (training code, configurations, and utilities): <https://github.com/AmerSaad001/Hybrid-Label-Noise-CV>.